

Lab 02 – Searching Algorithms

Task 01:

You have been assigned the task of creating a flavor search feature for a popular beverage brand, *Limca*. With around 15 popular Limca flavors available such as **Classic Lemon**, **Mint Splash**, **Citrus Zing**, **Lime & Ginger**, and **Masala Fizz**, the goal is to allow customers to easily search for specific Limca flavors and receive feedback on their availability.

If a flavor is found, display “**Flavor found**”. If the flavor isn’t found, notify the user that it’s **not currently offered**.

Code:

```
/*Limca flavor finder*/

#include<iostream>
#include<string>
using namespace std;

int main(){
    string flavors[5] = {"Classic Lemon", "Mint Splash", "Citrus Zing", "Lime &
Ginger", "Masala Fizz"};

    string targetFlavor;
    cout << "Enter flavor to find: ";
    getline(cin, targetFlavor);

    for(int i = 0; i < 5; i++){
        if(targetFlavor == flavors[i]){
            cout << "\nFlavor Found." << endl;
            return 0;
        }
    }
    cout << "\n" << targetFlavor << " is not currently offered." << endl;
}
```

Output:

```
Enter flavor to find: random

random is not currently offered.
PS D:\Hasan\DSA\University\Lab 02 - Searching Algorithms> ./a.exe
Enter flavor to find: Classic Lemon

Flavor Found.
```

Task 02:

Assume that you are a systems engineer aboard the space research vessel *Xenon Explorer*. During your expedition to a newly discovered planet, your team has detected a series of mysterious radio signal frequencies (in MHz) believed to be coming from ancient alien artifacts.

The artifact frequencies have been pre-recorded and sorted in ascending order:

103, 107, 112, 115, 118, 123, 130, 137, 145, 150, 158, 162, 170, 175, 180

Each frequency corresponds to a potential alien device emitting a unique signal. Some of them are harmless, but a few are believed to be high-energy devices that could interfere with the ship's systems, especially if their frequency is a multiple of 5. Your task is to write a C++ program that uses the binary search algorithm to quickly locate whether a specific frequency is part of the signal list.

- If the frequency is found and it's a **multiple of 5**, display:
"Warning: High-energy alien device detected at frequency X MHz!"
- If the frequency is found but not a multiple of 5, display:
"Signal found at frequency X MHz. Safe to investigate."
- If the frequency is **not found**, display:
"No such signal detected. Frequency X MHz not in range."

Code:

```
/*Alien Artifact Frequency Scanner*/

#include<iostream>
using namespace std;

int binarySearch(int arr[], int n, int target){
    int start = 0, end = n-1;

    while(start <= end){
        int mid = (start+end)/2;

        if(arr[mid] == target) return mid;
        else if(arr[mid] > target) end = mid-1;
        else start = mid+1;
    }
    return -1;
}

int main(){
    int freq[15] = {103, 107, 112, 115, 118, 123, 130, 137, 145, 150, 158, 162,
170, 175, 180};

    int targetFreq = 0;
```

```
    cout << "Enter the frequency to scan (in MHz): ";
    cin >> targetFreq;

    cout << "\n\nSearching..." << endl;
    int valueId = binarySearch(freq, 15, targetFreq);
    if(valueId != -1){
        if(freq[valueId]%5 == 0) cout << "Warning: High-energy alien device
detected at frequency "<<targetFreq<<" MHz!" << endl;
        else cout << "Signal found at frequency "<<targetFreq<<" MHz. Safe to
investigate" << endl;

        return 0;
    }
    cout << "No such signal detected. Frequency "<<targetFreq<<" MHz not in
range." << endl;
}
```

Output:

```
Enter the frequency to scan (in MHz): 230

Searching...
No such signal detected. Frequency 230 MHz not in range.
PS D:\Hasan\DSA\University\Lab 02 - Searching Algorithms> ./a.exe
Enter the frequency to scan (in MHz): 115

Searching...
Warning: High-energy alien device detected at frequency 115 MHz!
PS D:\Hasan\DSA\University\Lab 02 - Searching Algorithms> █
```

Task 03:

Welcome to the mystical realm of GemQuest! In this enchanting adventure, you are presented with the opportunity to embark on a gem hunt within the legendary treasure chest known as "GemChest." .The GemChest is said to hold a plethora of magical gems, each possessing unique properties. Begin your quest by visualizing the GemChest, a mystical container(of size 10) filled with dazzling gems(like Sapphire, ruby, diamond, emerald etc.). Ask the player to discover the number of occurrences of their chosen gem within the mystical GemChest. If found then display its occurrence count otherwise display not found. Write a C++ program for the above mystical GemChest.

Code:

```
/*Gem Hunt in a Treasure Chest*/

#include<iostream>
#include<string>
using namespace std;

int main(){
    string gemChest[10] = {"Sapphire", "Ruby", "Diamond", "Emerald", "Ruby",
"Topaz", "Sapphire", "Amethyst", "Diamond", "Ruby"};
    string targetGem;

    cout << "Welcome to GemQuest!" << endl;
    cout << "Enter the name of gem you seek: ";
    getline(cin, targetGem);

    int count = 0;
    for(int i = 0; i < 10; i++){
        if(gemChest[i] == targetGem) count++;
    }

    if(count > 0) cout << "\nYou found "<<count<<" "<<targetGem<<" in the
magical GemChest!" << endl;
    else cout << "\nNot found!" << endl;

}
```

Output:

```
Welcome to GemQuest!
Enter the name of gem you seek: Ruby

You found 3 Ruby in the magical GemChest!
```

Task 04:

Imtiaz Super Market is a well-known retail chain offering a wide range of household and grocery products to its customers. The supermarket maintains a detailed product database containing product names, sections, prices, and other relevant details. Your task is to implement a search functionality for Imtiaz Super Market's product system that allows customers to:

1. Search products by price
2. Search products by name

The system should display the product details such as product name, available section, and price when a match is found. If no product matches the user's search criteria, an appropriate message should notify the user that the product is not available.

Sample data:

Product Name	AvailableSections	Price (Rs.)
Tissue Roll	HouseholdSection	50.00
Soap	PersonalCareToiletriesSection	60.00
Toothpaste	ToiletriesSection	85.00
Bread	BakerySection	90.00
Sugar	GrocerySection	95.25

Code:

```
/*Product Lookup for Imtiaz Super Market*/

#include<iostream>
#include<string>
using namespace std;

struct product{
    string name;
    string section;
    float price;
};

// function to search products by name
void searchByName(product products[], int size, string targetName){
    bool found = false;

    for(int i = 0; i < size; i++){
        if(products[i].name == targetName){
            cout << "\nProduct Found!\n"
                << "Name: " << products[i].name << endl
                << "Section: " << products[i].section << endl
                << "Price: Rs. " << products[i].price << endl;
        }
    }
}
```

```
        found = true;
    }
}

if(!found){
    cout << "\nProduct named (" << targetName << ") not found." << endl;
}
}

// function to search products by price
void searchByPrice(product products[], int size, float targetPrice){
    bool found = false;

    for(int i = 0; i < size; i++){
        if(products[i].price == targetPrice){
            cout << "\nProduct Found!\n"
                << "Name: " << products[i].name << endl
                << "Section: " << products[i].section << endl
                << "Price: Rs. " << products[i].price << endl;
            found = true;
        }
    }

    if(!found){
        cout << "\nProduct with price (" << targetPrice << ") not found." <<
endl;
    }
}

int main(){
    product products[15] = {
        {"Tissue Roll", "HouseholdSection", 50.00},
        {"Soap", "PersonalCareToiletriesSection", 60.00},
        {"Toothpaste", "ToiletriesSection", 85.00},
        {"Bread", "BakerySection", 90.00},
        {"Sugar", "GrocerySection", 95.25},
        {"Milk", "DairySection", 120.50},
        {"Hand Sanitizer", "PersonalCareToiletriesSection", 150.00},
        {"Rice", "GrocerySection", 150.00},
        {"Eggs", "DairyPerishablesSection", 180.00},
        {"Tea", "GrocerySection", 200.00},
        {"Cooking Oil", "GroceryHouseholdSection", 230.00},
        {"Face Wash", "CosmeticsPersonalCare", 275.50},
        {"Cheese", "DairySection", 320.00},
        {"Shampoo", "PersonalCareToiletriesSection", 350.75},
```

```
        {"Washing Powder", "HouseholdCleaningSupplies", 400.00}
    };

    cout << "\n***Welcome to Imtiaz Super Market***" << endl << endl;

    int choice = 0;
    do{
        cout << "\n1. Search products by name." << endl;
        cout << "2. Search products by price." << endl;
        cout << "3. Exit." << endl;
        cout << "Enter choice: ";
        cin >> choice;
        cin.ignore();

        // search by name
        if(choice == 1){
            string targetName;
            cout << "\nEnter the name of product to find: ";
            getline(cin, targetName);
            searchByName(products, 15, targetName);
        }

        // search by price
        else if(choice == 2){
            float targetPrice = 0.0;
            cout << "\nEnter the price of product to find: ";
            cin >> targetPrice;
            searchByPrice(products, 15, targetPrice);
        }

        else if(choice == 3){
            cout << "\nThank you for shopping :)" << endl;
        }

        else{
            cout << "\nInvalid choice!" << endl;
        }

    }while(choice != 3);
}
```

Output:

```
***Welcome to Imtiaz Super Market***

1. Search products by name.
2. Search products by price.
3. Exit.
Enter choice: 1

Enter the name of product to find: Random

Product named (Random) not found.

1. Search products by name.
2. Search products by price.
3. Exit.
Enter choice: 2

Enter the price of product to find: 200

Product Found!
Name: Tea
Section: GrocerySection
Price: Rs. 200

1. Search products by name.
2. Search products by price.
3. Exit.
Enter choice: 3

Thank you for shopping :)
```